

Chapter 7

Attitude Control

Attitude control means to use the measurements and a reference to calculate a torque demand that will make the measured state equal to the reference. Following the argument in [5, (pp154-156)], that the satellite body coordinate system and a target coordinate system will be aligned when zero rotation is needed to align the satellite body system with the axes of the target system. This can be done using a control law

$$\mathbf{u} = -\mathbf{K}_p \mathbf{q}_e - \mathbf{K}_{pd} \boldsymbol{\omega} \quad (7.1)$$

where $\mathbf{K}_p$ and $\mathbf{K}_{pd}$ are controller gain matrices, $\boldsymbol{\omega}$ the angular velocity vector of the satellite and $\mathbf{q}_e$ is the quaternion error,

$$\begin{bmatrix} q_{1e} \\ q_{2e} \\ q_{3e} \\ q_{4e} \end{bmatrix} = \begin{bmatrix} q_{4r} & q_{3r} & -q_{2r} & -q_{1r} \\ -q_{3r} & q_{4r} & q_{1r} & q_{2r} \\ q_{2r} & -q_{1r} & q_{4r} & -q_{3r} \\ q_{1r} & q_{2r} & q_{3r} & q_{4r} \end{bmatrix} \begin{bmatrix} q_{1s} \\ q_{2s} \\ q_{3s} \\ q_{4s} \end{bmatrix} \quad (7.2)$$

i.e. a quaternion that describes the rotation needed to align the satellite body coordinate axes with those of the target coordinate system. The quaternion $\mathbf{q}_s$ describes the rotation to go from inertial axes to satellite body coordinate system axes, $\mathbf{q}_r$ the rotation of inertial axes to the reference (target) coordinate system axes.

The feedback in Eq. 7.1 has $\mathbf{q}_t$ as reference for pointing, but no reference in $\boldsymbol{\omega}$. When we want to make a reorientation manoeuvre with a spacecraft, it could be convenient to introduce a reference also in $\boldsymbol{\omega}$, to make a manoeuvre that can follow an angular rate, e.g. when a satellite shall track another object or a point on Earth. A control law with reference in both quaternion and in angular velocity is:

$$\mathbf{u} = -\mathbf{K}_p \mathbf{q}_e - \mathbf{K}_{pd} \boldsymbol{\omega}_e \quad (7.3)$$

where

$$\boldsymbol{\omega}_e = \boldsymbol{\omega}_s - \boldsymbol{\omega}_r \quad (7.4)$$

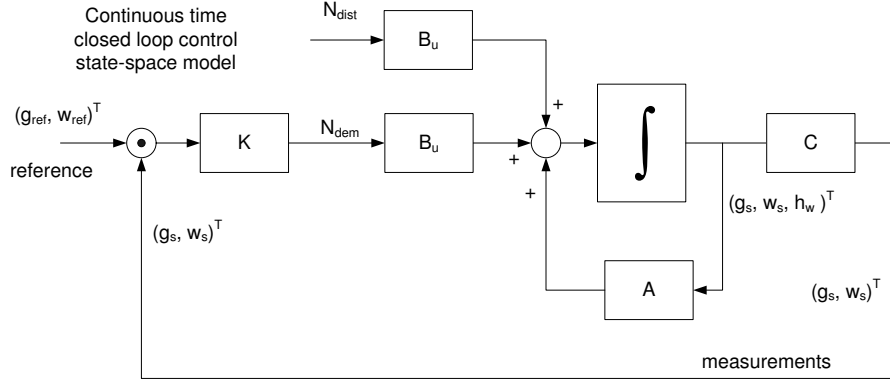


Figure 7.1: Closed loop with output feedback and reference to the loop from both ω and $\mathbf{g}$ in the linear state space representation of the satellite.

We will analyse both cases in the sequel, Eq. 7.3 with reference values in both angular velocity and attitude, Eq. 7.1 with reference quaternion but zero angular velocity as reference.

7.1 Reference in both angular rate and attitude

Attitude control means to use the measurements and a reference from both angular rate and direction (quaternion) to obtain a closed loop control as illustrated in Figure 7.1

The feedback law is taken to be

$$\mathbf{N}_d = -\mathbf{K}_p \begin{bmatrix} q_{1e} q_{4e} \\ q_{2e} q_{4e} \\ q_{3e} q_{4e} \end{bmatrix} - \mathbf{K}_{pd} \begin{bmatrix} \omega_{1e} \\ \omega_{2e} \\ \omega_{3e} \end{bmatrix} \quad (7.5)$$

where the quaternion error is defined as

$$\mathbf{q}_e = \mathbf{q}_s^{-1} \mathbf{q}_r \quad (7.6)$$

and the angular velocity error is

$$\omega_e = \omega_s - \omega_r \quad (7.7)$$

For small angles, and taking q_4 positive,

$$\mathbf{q}_e \simeq \begin{bmatrix} g_{1r} - g_{1s} \\ g_{2r} - g_{2s} \\ g_{3r} - g_{3s} \\ 1 \end{bmatrix}. \quad (7.8)$$

and the control law can be written as

$$\begin{aligned} \mathbf{N}_d &= \mathbf{K}_p \begin{bmatrix} g_{1r} - g_{1s} \\ g_{2r} - g_{2s} \\ g_{3r} - g_{3s} \end{bmatrix} + \mathbf{K}_{pd} \begin{bmatrix} \omega_{1r} - \omega_{1s} \\ \omega_{2r} - \omega_{2s} \\ \omega_{3r} - \omega_{3s} \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{K}_{pd} & \mathbf{K}_p \end{bmatrix} \begin{bmatrix} \omega_{1r} - \omega_{1s} \\ \omega_{2r} - \omega_{2s} \\ \omega_{3r} - \omega_{3s} \\ g_{1r} - g_{1s} \\ g_{2r} - g_{2s} \\ g_{3r} - g_{3s} \end{bmatrix} \end{aligned} \quad (7.9)$$

The linear model has the state vector $x = [\omega_1, \omega_2, \omega_3, g_1, g_2, g_3, h_1, h_2, h_3]^T$ and the measurement is $y = [\omega_1, \omega_2, \omega_3, g_1, g_2, g_3]^T$. With reference in both ω and $\mathbf{g}$, $\mathbf{y}_{ref} = [[\omega_{1r}, \omega_{2r}, \omega_{3r}, g_{1r}, g_{2r}, g_{3r}]^T]$ y this is the same as

$$\begin{aligned} \mathbf{N}_d &= \begin{bmatrix} \mathbf{K}_{pd} & \mathbf{K}_p \end{bmatrix} (\mathbf{y}_r - \mathbf{y}) \\ &= - \begin{bmatrix} \mathbf{K}_{pd} & \mathbf{K}_p \end{bmatrix} \mathbf{C}\mathbf{x} + \begin{bmatrix} \mathbf{K}_{pd} & \mathbf{K}_p \end{bmatrix} \mathbf{y}_r \\ &= -\mathbf{K}_y \mathbf{C}\mathbf{x} + \mathbf{K}_r \mathbf{y}_r \end{aligned} \quad (7.10)$$

where the last two lines define the gain matrices $\mathbf{K}_y = \begin{bmatrix} \mathbf{K}_{pd} & \mathbf{K}_p \end{bmatrix}$ and $\mathbf{K}_r = \begin{bmatrix} \mathbf{K}_{pd} & \mathbf{K}_p \end{bmatrix}$.

The transfer function from reference to output with the closed feedback is then

$$\mathbf{y}(s) = \mathbf{C}(s\mathbf{I}_{9,9} - (\mathbf{A} - \mathbf{B}_u \mathbf{K}_y \mathbf{C}))^{-1} \mathbf{B}_u \mathbf{K}_r \mathbf{y}_r(s) \quad (7.11)$$

The form of this expression is

$$\mathbf{y}(s) = \mathbf{C}(s\mathbf{I}_{9,9} - \mathbf{A}_{cl})^{-1} \mathbf{B}_{cl} \mathbf{y}_r(s) \quad (7.12)$$

with

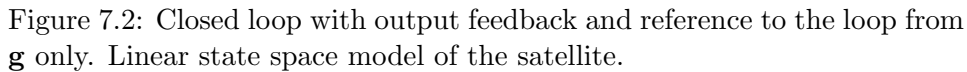
$$\mathbf{A}_{cl} = (\mathbf{A} - \mathbf{B}_u \mathbf{K}_y \mathbf{C}) \quad (7.13)$$

$$\mathbf{B}_{cl} = \mathbf{B}_u \mathbf{K}_r = \mathbf{B}_u \begin{bmatrix} \mathbf{K}_{pd} & \mathbf{K}_p \end{bmatrix} \quad (7.14)$$

$$\mathbf{C}_{cl} = \mathbf{C} \quad \text{and} \quad \mathbf{D}_{cl} = \mathbf{0}_{6,6} \quad (7.15)$$

The closed loop transfer functions 7.12 are calculated using the commonplace Matlab functions (see below), using the LTI system definition

$$sys_{cl} = ss(\mathbf{A}_{cl}, \mathbf{B}_{cl}, \mathbf{C}_{cl}, \mathbf{D}_{cl}). \quad (7.16)$$



When the reference is not desired in angular velocity, the block diagram 7.2 is applicable.

$$\begin{aligned} \mathbf{N}_d &= - \begin{bmatrix} \mathbf{K}_{pd} & \mathbf{K}_p \end{bmatrix} \mathbf{y} + \mathbf{K}_p \mathbf{g}_r \\ &= - \begin{bmatrix} \mathbf{K}_{pd} & \mathbf{K}_p \end{bmatrix} \mathbf{C} \mathbf{x} + \mathbf{K}_p \mathbf{g}_r \\ &= - \mathbf{K}_y \mathbf{C} \mathbf{x} + \mathbf{K}_r \mathbf{g}_r \end{aligned} \quad (7.17)$$
$$\mathbf{y}(s) = \mathbf{C}(\mathbf{sI}_{9,9} - (\mathbf{A} - \mathbf{B}_u\mathbf{K}_y\mathbf{C}))^{-1}\mathbf{B}_u\mathbf{K}_r\mathbf{g}_r(s) \quad (7.18)$$
$$\mathbf{A}_{cl} = (\mathbf{A} - \mathbf{B}_u \mathbf{K}_y \mathbf{C}) \quad (7.19)$$

$$\mathbf{B}_{cl} = \mathbf{B}_u \mathbf{K}_r = \mathbf{B}_u \mathbf{K}_p \quad (7.20)$$

$$\mathbf{C}_{cl} = \mathbf{C} \quad \text{and} \quad \mathbf{D}_{cl} = \mathbf{0}_{6,3} \quad (7.21)$$

$$\mathbf{y}(s) = \mathbf{C}(s\mathbf{I}_{9,9} - \mathbf{A}_{cl})^{-1}\mathbf{B}_{cl}\mathbf{y}_r(s) \quad (7.22)$$

7.3 Stability Analysis and Simple Controller Design

The essential properties of a closed loop control system include closed loop eigenvalues to analyse stability, bode plot from reference input to output, step response to unit step at the reference input, and response to torque disturbances.

Design of closed loop control systems can be achieved in several elegant ways, including optimal and robust control methods. Even with a limited background in control theory, one can design well performing controllers using the stepwise procedure below.

Algorithm 1 *Stepwise procedure for simple control system design*

Commands are Matlab® control systems toolbox and the supplied functions for the course.

1. Determine the linear model without wheel dynamics using
 $[satmod_c, satmod_dist] = lin_sat_wheels_mod(Is, ws, hw)$
2. Eigenvalues in the open loop (no control) system using
 $damp(satmod_c)$
 $pzmap(satmod_c)$
3. Get the $\mathbf{A}, \mathbf{B}_u, \mathbf{C}, \mathbf{D}$ matrices using
 $[\mathbf{A}, \mathbf{B}_u, \mathbf{C}, \mathbf{D}] = ssdata(satmod_c)$
4. Get the $\mathbf{A}, \mathbf{B}_{dist}, \mathbf{C}, \mathbf{D}$ matrices using
 $[\mathbf{A}, \mathbf{B}_d, \mathbf{C}, \mathbf{D}] = ssdata(satmod_dist)$
5. Define the closed loop by
 $\mathbf{D}_{cl} = zeros(size(\mathbf{C}, 1), size(\mathbf{K}_r, 2))$
 $[satmod_c_closed] = ss((\mathbf{A} - \mathbf{B}_u \mathbf{K}_y \mathbf{C}, \mathbf{B}_u \mathbf{K}_r, \mathbf{C}_{cl}, \mathbf{D}_{cl}))$
6. Investigate eigenvalues of the closed loop by
 $damp(satmod_c_closed)$
 $pzmap(satmod_c_closed)$ and
 $[P, Z] = pzmap(satmod_c_closed)$
7. Once a suitable stable controller has been found, determine frequency response properties by
 $bode(satmod_c_closed)$
8. and step responses by
 $step(satmod_c_closed)$

9. Investigate disturbance rejection by
 $[satmod_dist_closed] = ss((\mathbf{A} - \mathbf{B}_u \mathbf{K}_y \mathbf{C}, \mathbf{B}_d, \mathbf{C}, \mathbf{D}))$
 $bode(ssatmod_dist_closed)$
 $step(satmod_dist_closed)$
10. iterate in the above procedure until specifications are met.

Example 3 Rømer-like Satellite

Let the Rømer-like satellite have

$$\boldsymbol{\omega} = (0, 0, 0)^T [\text{rad/s}]; \quad h_i = [0, 0.5, 0.2]^T [\text{Nms}]; \quad (7.23)$$

$$\mathbf{I}_s = \begin{bmatrix} 14.3 & 0 & 0 \\ 0 & 13.6 & 0 \\ 0 & 0 & 4.6 \end{bmatrix} [\text{kgm}^2] \quad (7.24)$$

Define the desired bandwidth of the closed-loop system by $\omega_{bw_cls} = 0.2$ [rad/s]. A simple design approach is to select gains to be

$$\begin{aligned} k_p(j) &= \mathbf{I}_s(j, j) \omega_{bw_cls}^2 \\ k_{pd}(j) &= 2\mathbf{I}_s(j, j) \omega_{bw_cls} \end{aligned} \quad (7.25)$$

$$\begin{aligned} \mathbf{K}_p &= \text{diag}(k_p(1), k_p(2), k_p(3)) \\ \mathbf{K}_{pd} &= \text{diag}(k_{pd}(1), k_{pd}(2), k_{pd}(3)) \end{aligned} \quad (7.26)$$

We investigate the properties of this simplistic design idea by using the steps of algorithm 1. The gain factors used are the above given $\mathbf{K}_p$ and $\mathbf{K}_{pd}$. Combinations of two sets of nominal values are used, $\{\omega_s = [0; 0; 0], \omega_s = [0; -0.003; 0]\}$ and $\{\mathbf{h}_s = [0; 0; 0], \mathbf{h}_s = [0; 0; 0.5]\}$. The results are shown in Figures 7.3, 7.5 and 7.7.

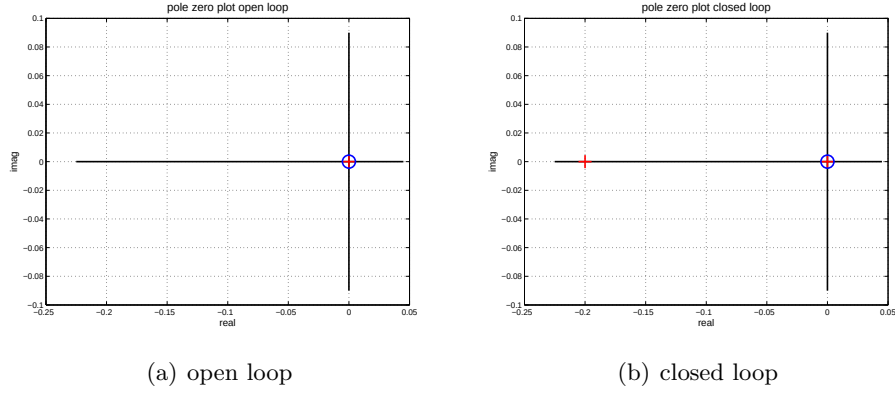


Figure 7.3: R loop (b). Angular velocity is $\omega_s = [0; 0; 0]$ rad/s and angular momentum is $\mathbf{h}_s = [0; 0; 0]$ Nms. The closed loop eigenvalues are all in -0.2 as intended. The small imaginary part is due to numerics. Three eigenvalues at the origin are cancelled by three zeros. The origin of this is the physical integration from $\dot{h}$ to h .

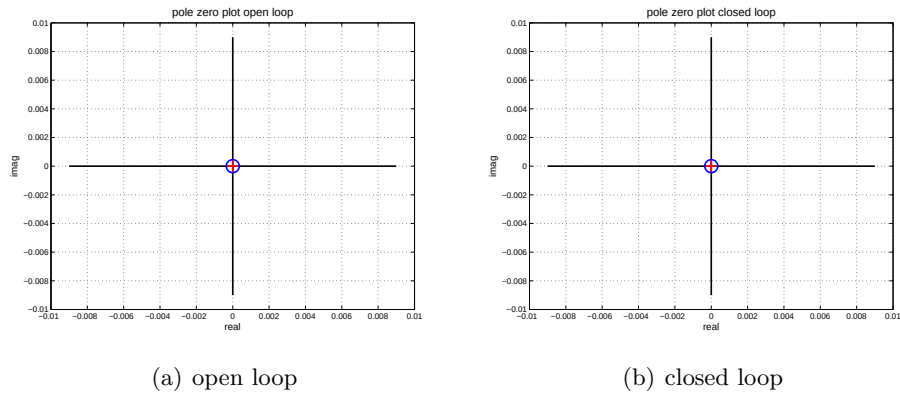


Figure 7.4: Same as Figure 7.3 but zoomed

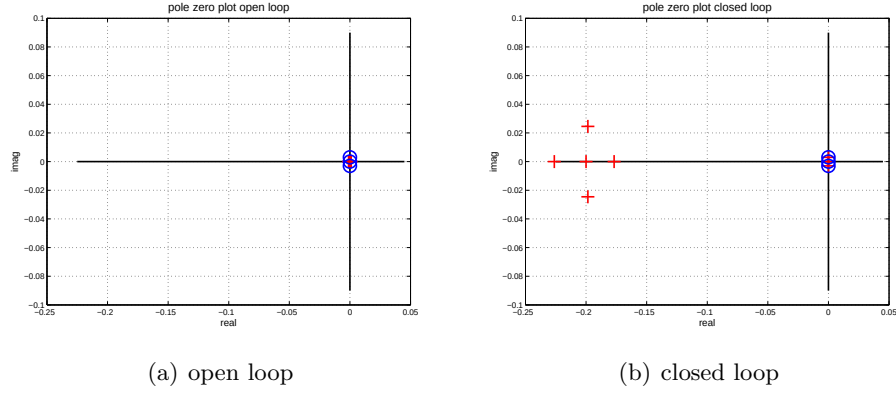


Figure 7.5: Roemer-like satellite pole-zero plot for open (a) and closed loop (b). Angular velocity is $\omega_s = [0; -0.003; 0]$ rad/s and angular momentum $\mathbf{h}_s = [0; 0; 0.5]$ Nms. The angular velocity around the y-axis, and the moment of inertia with $I_x < I_y < I_z$, cause the open loop system to be unstable. The closed loop is stable with closed loop poles only slightly affected by the perturbation of the open loop system.

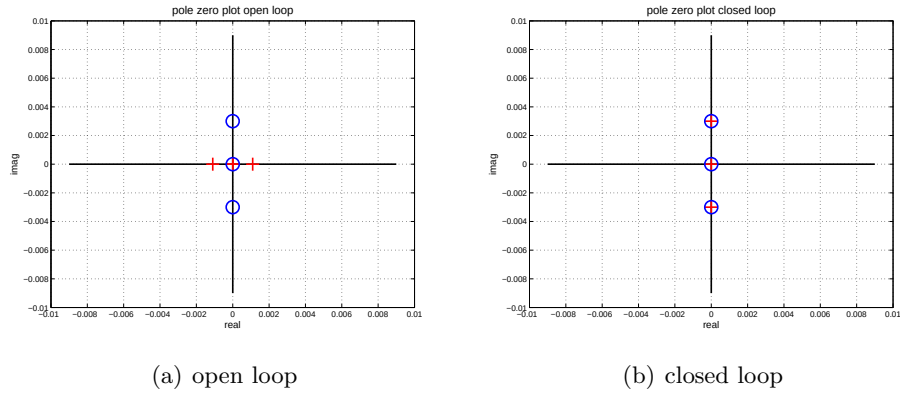


Figure 7.6: Same as Figure 7.5 but zoomed

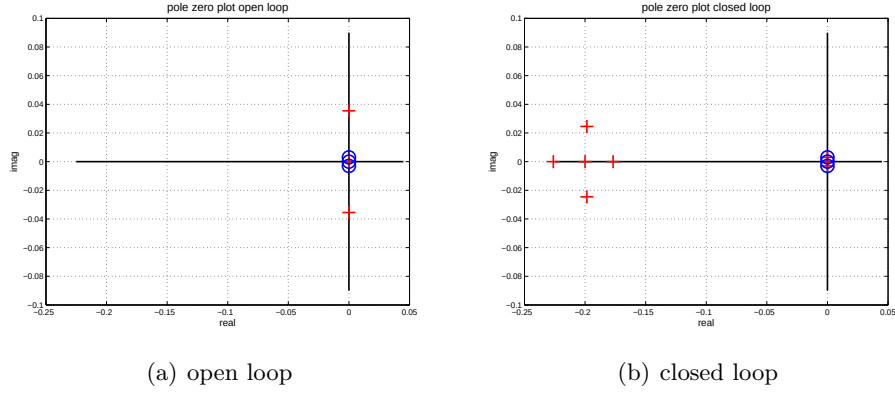


Figure 7.7: Roemer-like satellite pole-zero plot for open (a) and closed loop (b). Angular velocity is $\omega_s = [0; -0.003; 0]$ rad/s and angular momentum $\mathbf{h}_s = [0; 0; 0.5]$ Nms. The bias in angular momentum has moved the open loop pole-zero pattern significantly. Despite the large perturbation, the closed loop poles show that the closed loop system is still nicely damped.

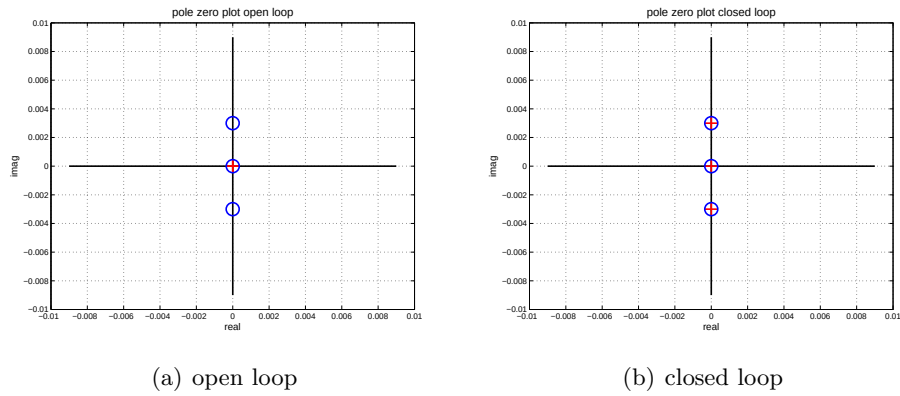


Figure 7.8: Same as Figure 7.7 but zoomed

Bibliography

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